

Sangeek Hyun

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RESEARCH SUMMARY

I am a Ph.D. candidate in Artificial Intelligence at Sungkyunkwan University, working at the intersection of generative modeling, computer vision, and visual representation learning. My research aims to build scalable generative systems that can model, synthesize, and manipulate visual worlds across images, videos, and 3D scenes. I am particularly interested in how generative models can learn spatial structure, temporal dynamics, and efficient representations for high-fidelity and controllable visual generation.

RESEARCH INTERESTS

- Generative Models, Visual World Modeling, Video and 3D Generation, 3D Scene Representation, Efficient and Scalable Generation.

EDUCATION

Sungkyunkwan University (SKKU) Ph.D. Candidate, Artificial Intelligence (Visual Computing Lab) • Advisor: Prof. Jae-Pil Heo	Suwon, South Korea <i>Sep. 2021 – Present</i>
Sungkyunkwan University (SKKU) M.S. in Artificial Intelligence (Visual Computing Lab) • Advisor: Prof. Jae-Pil Heo	Suwon, South Korea <i>Feb. 2020 – Aug. 2021</i>
Sungkyunkwan University (SKKU) B.S. in Computer Science and Engineering	Suwon, South Korea <i>Mar. 2016 – Feb. 2020</i>

HONORS AND AWARDS

- **Winner**, Qualcomm Innovation Fellowship Korea (QIFK) 2024

PUBLICATIONS

Cross-scale Aligned Supervision for Training GANs <u>Sangeek Hyun</u> , MinKyu Lee, Jae-Pil Heo arXiv preprint	2026
Disambiguating 2D-3D Correspondences in Gaussian Splatting-based Feature Fields for Visual Localization Miso Lee, <u>Sangeek Hyun</u> , Yerim Jeon, Jae-Pil Heo arXiv preprint	2026
Scalable GANs with Transformers <u>Sangeek Hyun</u> , MinKyu Lee, Jae-Pil Heo <i>International Conference on Machine Learning (ICML)</i>	2026
Scalable GANs with Transformers <u>Sangeek Hyun</u> , MinKyu Lee, Jae-Pil Heo <i>International Conference on Machine Learning (ICML)</i>	2026
Scalable GANs with Transformers <u>Sangeek Hyun</u> , MinKyu Lee, Jae-Pil Heo <i>International Conference on Machine Learning (ICML)</i>	2026

SeaCache: Spectral-Evolution-Aware Cache for Accelerating Diffusion Models

Jiwoo Chung, [Sangeek Hyun](#), MinKyu Lee, Byeongju Han, Geonho Cha, Dongyoon Wee, Youngjun Hong, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) **[Oral]** 2026

Looking Beyond the Window: Global-Local Aligned CLIP for Training-free Open-Vocabulary Semantic Segmentation

ByeongCheol Lee, Hyun Seok Seong, [Sangeek Hyun](#), Gilhan Park, WonJun Moon, and Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026

Rethinking LayerNorm in Image Restoration Transformers

MinKyu Lee, [Sangeek Hyun](#), Woojin Jun, Hyunjun Kim, Jiwoo Chung, Jae-Pil Heo

International Conference on Learning Representations (ICLR) 2026

Correlation-guided calibration of query dependency for video temporal grounding

WonJun Moon, [Sangeek Hyun](#), SuBeen Lee, Jae-Pil Heo

Pattern Recognition PR 2025

Diffusion Feature Field for Text-based 3D Editing with Gaussian Splatting

Eunseo Koh, [Sangeek Hyun](#), MinKyu Lee, Jiwoo Chung, Kangmin Seo, Jae-Pil Heo

Advances in Neural Information Processing Systems (NeurIPS) 2025

Fine-Tuning Visual Autoregressive Models for Subject-Driven Generation

Jiwoo Chung, [Sangeek Hyun](#), Hyunjun Kim, Eunseo Koh, MinKyu Lee, Jae-Pil Heo

International Conference on Computer Vision (ICCV) 2025

AESOP: Auto-Encoded Supervision for Perceptual Image Super-Resolution

MinKyu Lee, [Sangeek Hyun](#), Woojin Jun, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025

GSGAN: Adversarial Learning for Hierarchical Generation of 3D Gaussian Splats

[Sangeek Hyun](#), Jae-Pil Heo

Advances in Neural Information Processing Systems (NeurIPS) 2024

Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer

Jiwoo Chung*, [Sangeek Hyun](#)*, Jae-Pil Heo (*Equal contribution)

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) **[Highlight]** 2024

Diversity-aware Channel Pruning for StyleGAN Compression

Jiwoo Chung, [Sangeek Hyun](#), Sang-Heon Shim, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2024

Task-disruptive Background Suppression for Few-Shot Segmentation

Suho Park, SuBeen Lee, [Sangeek Hyun](#), Hyun Seok Seong, Jae-Pil Heo

AAAI Conference on Artificial Intelligence (AAAI) 2024

Frequency-based Motion Representation for Video Generative Adversarial Networks

[Sangeek Hyun](#), Jaihyun Lew, Jiwoo Chung, Euiyeon Kim, Jae-Pil Heo

IEEE Transactions on Image Processing (TIP) 2023

Disentangled Representation Learning for Unsupervised Neural Quantization

Haechan Noh, [Sangeek Hyun](#), Woojin Jeong, Hanshin Lim, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2023

Query-dependent Video Representation for Moment Retrieval and Highlight Detection

WonJun Moon*, Sangeek Hyun*, SangUk Park, Dongchan Park, Jae-Pil Heo (***Equal contribution**)

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2023

Local Attention Pyramid for Scene Image Generation

Sang-Heon Shim, Sangeek Hyun, DaeHyun Bae, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2022

Self-Supervised Video GANs: Learning for Appearance Consistency and Motion Coherency

Sangeek Hyun, Jihwan Kim, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2021

VarSR: Variational Super-Resolution Network for Very Low Resolution Images

Sangeek Hyun, Jae-Pil Heo

European Conference on Computer Vision (ECCV)

2020

ACADEMIC SERVICES

Conference Reviewer: CVPR (2022–2026), ICCV (2025), ECCV (2024, 2026), NeurIPS (2024–2026), ICLR (2025–2026), ICML (2025–2026), AAAI (2026)

Journal Reviewer: IJCV (2025–2026)

PROJECTS

Named Entity Recognition for Video Understanding

Apr. 2025 – Mar. 2026

- Developed video NER models for entity extraction and video understanding using multimodal large language models, and built evaluation pipelines for systematic benchmarking.

Detection of AI-Generated Visual Content

Jul. 2021 – Feb. 2025

- Co-led a project on AI-generated visual content detection, including dataset construction and the development of deepfake and synthetic media detection models. * *Recognized by the Minister of Science and ICT for outstanding achievement in social problem-solving R&D.*

Fashion Image Retrieval and Generation

Sep. 2020 – Dec. 2021

- Developed models for fashion image retrieval and generation, focusing on image synthesis and visual feature learning for fashion applications.

Handwritten Mathematical Expression Recognition

Mar. 2020 – Dec. 2020

- Developed handwritten mathematical expression recognition models and built paired datasets of handwritten formula images and LaTeX annotations.

Semiconductor Defect Detection and Classification

Sep. 2019 – Feb. 2020

- Developed computer vision models for semiconductor defect detection and classification with an emphasis on robust visual inspection in few-shot industrial data.